



Barker

College

2024

YEAR 12

ASSESSMENT TASK 3

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Student Number

Mathematics Advanced

Staff Involved:

- PCB* • AXD • GPF • ECB • EJK
- RAS • DOB • JAI • SJB

Wednesday 5th June, 8:30am

145 copies

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General Instructions

- Working time – 50 minutes
- Write using blue or black pen
- Write your Barker Student number in the spaces indicated
- Write your answers in the spaces provided
- Board approved calculators may be used
- The HSC Reference Sheet is provided.
- Attempt all questions
- ALL necessary working must be shown
- Marks may not be awarded for careless or badly arranged work
- Diagrams are NOT to scale

Topics Covered

- Exponential and Logarithmic Functions
- Trigonometric Functions

Section	Marks
Question 1	/ 11
Question 2	/ 12
Question 3	/ 8
Question 4	/ 9
Total Marks:	/ 40

Question 1 (11 marks)

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Student number

(a) Differentiate with respect to x :

(α) $y = 5 \tan\left(\frac{x}{4}\right)$ 1

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(β) $y = e^{3x^3+5}$ 1

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(γ) $y = 3x^2 \ln(2x)$, fully simplify your answer. 2

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(b) Find the following indefinite integrals:

(α) $\int \frac{7}{3x+1} dx$ 2

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(β) $\int 3 \sin(\pi x) dx$ 2

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(c) Evaluate, fully simplifying your answer: 3

$\int_{\ln 2}^{\ln 3} e^{3x} + 2e^x dx$

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End of Question 1

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Question 4 (9 marks)

Student number

(a) Consider the function $y = \sec x$

(i) Show, with clear working, that $\frac{dy}{dx} = \sec x \tan x$

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(ii) Hence, find $\int \tan x (1 + \sec x) dx$

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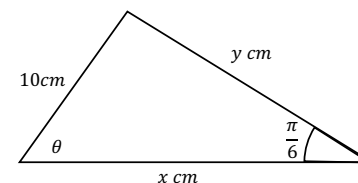
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(b) A triangle has a side length of 10 cm and angle of $\frac{\pi}{6}$ radians as shown.



(i) Show that the length of y can be written as $y = 20 \sin \theta$

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(ii) Hence, show that the perimeter, P , of the triangle can be written as

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$$P = 10 + 20 \sin \theta + 20 \sin\left(\frac{5\pi}{6} - \theta\right)$$

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Question 4(b) continues on page 13

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Student number

(iii) Find the value of θ that maximises the perimeter.

3

Year 12 Mathematics Advanced - Assessment 3 - 2024 - Solutions

Q1 a)

$$\alpha) y' = \frac{5}{4} \sec^2\left(\frac{x}{4}\right)$$

$$\beta) y' = (9x^2) e^{3x^3+5}$$

$$\gamma) y' = (6x)(\ln 2x) + \left(\frac{2}{2x}\right)(3x^2) \\ = 6x \ln 2x + 3x$$

b)

$$\alpha) \frac{7}{3} \int \frac{3}{3x+1} dx \\ = \frac{7}{3} \ln|3x+1| + C$$

$$\beta) \frac{3}{\pi} \int \pi \sin(\pi x) dx \\ = -\frac{3}{\pi} \cos(\pi x) + C$$

$$\gamma) \frac{1}{3} \int_{\ln 2}^{\ln 3} 3e^{3x} dx + 2 \int_{\ln 2}^{\ln 3} e^x dx$$

$$= \frac{1}{3} [e^{3x}]_{\ln 2}^{\ln 3} + 2 [e^x]_{\ln 2}^{\ln 3}$$

$$= \frac{1}{3} (e^{3 \ln 3} - e^{3 \ln 2}) + 2 (e^{\ln 3} - e^{\ln 2})$$

$$= \frac{25}{3}$$

Q2 a)

$$g(x) = \int e^{2x-6} + x dx \\ = \frac{1}{2} \int 2e^{2x-6} dx + \int x dx \\ = \frac{1}{2} e^{2x-6} + \frac{x^2}{2} + C$$

$$4 = \frac{1}{2} e^0 + \frac{9}{2} + C$$

$$-1 = C$$

$$g(x) = \frac{1}{2} e^{2x-6} + \frac{x^2}{2} - 1$$

$$b) y' = 3 \cos x \sin^2 x \\ m_T = 3 \cos\left(\frac{\pi}{3}\right) \sin^2\left(\frac{\pi}{3}\right) \\ = \frac{9}{8} \\ m_N = -\frac{8}{9}$$

c)

$$i) 0 = \frac{6}{x} - 3$$

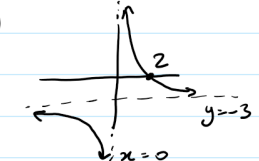
$$3 = \frac{6}{x}$$

$$x = \frac{6}{3}$$

$$x = 2$$

$$\therefore (2, 0)$$

ii)



$$iii) A = \int_1^2 \frac{6}{x-3} dx + \left| \int_2^5 \frac{6}{x-3} dx \right|$$

$$= [6 \ln|x-3|]_1^2 + \left| [6 \ln|x-3|]_2^5 \right|$$

$$= (6 \ln 2 - 6) - (6 \ln 1 - 3) + (6 \ln 5 - 15) - (6 \ln 2 - 6)$$

$$= 4.66 \text{ units}^2$$

End of Question 4

End of Assessment

Q3 a)

i) $y = 8 \ln x - 6x + \frac{x^2}{2}$
 $y' = \frac{8}{x} - 6 + x$

ii) stat pts when $y' = 0$

$$0 = \frac{8}{x} - 6 + x$$

$$0 = x^2 - 6x + 8$$

$$0 = (x-4)(x-2)$$

$$x = 2, 4$$

x	1	2	3	4	5
y'	3	0	$-\frac{1}{3}$	0	$\frac{3}{5}$
	/	-	\	-	/

$$x = 2 \text{ max TP}$$

$$x = 4 \text{ min TP}$$

b) $A = \int_0^1 \ln 8(2^x) dx - \int_0^1 \ln 3(3^x) dx$

$$= \left[\ln 8 \left(\frac{2^x}{\ln 2} \right) \right]_0^1 - \left[\ln 3 \left(\frac{3^x}{\ln 3} \right) \right]_0^1$$

$$= \frac{\ln 8}{\ln 2} (2-1) - (3-1)$$

$$= 1$$

Q4 a)

i) $y = \frac{1}{\cos x} = (\cos x)^{-1}$

$$y' = (-1)(-\sin x)(\cos x)^{-2}$$

$$= \frac{\sin x}{\cos^2 x}$$

$$= \frac{1}{\cos x} \times \frac{\sin x}{\cos x}$$

$$= \sec x \tan x$$

ii) $\int \tan x dx + \int \sec x \tan x dx$

$$= -\int \frac{\sin x}{\cos x} dx + \sec x + C$$

$$= -\ln |\cos x| + \sec x + C$$

b) i) $\frac{y}{\sin \theta} = \frac{10}{\sin \frac{\pi}{6}}$

$$y = \frac{10 \sin \theta}{\frac{1}{2}}$$

$$y = 20 \sin \theta$$

ii) $\frac{x}{\sin(\pi - \frac{\pi}{6} - \theta)} = \frac{10}{\sin \frac{\pi}{6}}$

$$x = 20 \sin \left(\frac{5\pi}{6} - \theta \right)$$

$$p = 10 + y + x$$

$$= 10 + 20 \sin \theta + 20 \sin \left(\frac{5\pi}{6} - \theta \right)$$

iii) $p' = 20 \cos \theta - 20 \cos \left(\frac{5\pi}{6} - \theta \right)$

possible max when $p' = 0$

$$\cos \left(\frac{5\pi}{6} - \theta \right) = \cos \theta$$

$$\frac{5\pi}{6} - \theta = \theta$$

$$\frac{5\pi}{6} = 2\theta$$

$$\frac{5\pi}{12} = \theta$$

θ	1.3	$\frac{5\pi}{12}$	1.4
p'	0.34	0	-3.51
	/	-	\

$$\therefore \text{max at } \theta = \frac{5\pi}{12}$$